Supplementary Material

Exact Rational Certificate for $(405, 132, 63, 33)$

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14 October 2026

S1 Complete rational certificate for 405

This supplement specifies all 174 sources and their coefficients for $(405, 132, 63, 33)$, including every value of all 107 weight functions. It also gives formulas that recover every coefficient of the contradiction from this printed data. No numerical optimizer is part of this specification.

A graph in the tables is specified by (h, b): it has vertex set $\{0, \dots, h-1\}$ and an edge ij, $i < j$, exactly when bit $j(j-1)/2 + i$ of the nonnegative integer b is one. The symbol y_F means $2^{\binom{h}{2}} N_F/(405)_h$, where $h = |F|$, and is unchanged by relabelling F. The source index i runs from 1 to 174. The printed coefficient w_i is already reduced; no further row normalization is applied.

Let Γ be a hypothetical strongly regular graph with parameters $(405, 132, 63, 33)$. For a labelled graph F, let N_F count its injective induced embeddings into Γ. Write $(405)_h = \prod_{j=0}^{h-1}(405 - j)$, with $(405)_0 = 1$, and $[m] = \{0, \dots, m-1\}$. For a graph H on $[m]$, $\mathcal{N}_H(b)$ is the neighbourhood of b, and H_C adds one labelled vertex adjacent precisely to $C \subseteq [m]$. A binary profile has coordinate j equal to one precisely when its vertex is adjacent to selected vertex j. The SRG identities give the positive semidefinite matrix

$$G = \frac{9I + 3\,\mathrm{Adj}(\Gamma) - J}{8}, \qquad \operatorname{rank} G = 30.$$

Here J is the all-one matrix. Thus G is the Gram matrix of unit vectors $p_x \in \mathbb{R}^{30}$.

Exact reconstruction formulas

For $i \le 67$ set $R_i(y) = -y_{F_i}$ using Table S1. For $i \ge 68$, take H_i, ℓ_i and f_i from Table S2. An array in that table is read row by row in binary lexicographic profile order, with coordinate 0 first. Thus the first entry is $f_i(0, \dots, 0)$ and the last is $f_i(1, \dots, 1)$. All $2^{|H_i|}$ entries are present.

For an m-vertex anchor H and profiles $\sigma, \tau \in \{0,1\}^m$, put

$$\begin{aligned}
G_H &= (9I + 3\,\mathrm{Adj}(H) - J)/8, & b_\sigma &= (3\sigma - \mathbf{1})/8,\\
h_\sigma &= 1 - b_\sigma^T G_H^{-1} b_\sigma, & z_{\sigma\tau}(e) &= (3e-1)/8 - b_\sigma^T G_H^{-1} b_\tau,\\
D &= 30 - m, & (\kappa_0, \kappa_1, \kappa_2)(z, h, h') &= \left(1, z, \frac{Dz^2 - hh'}{D-1}\right).
\end{aligned}$$

Here H_σ adds one labelled vertex with profile σ, and $H_{\sigma,\tau;e}$ adds two labelled vertices with those profiles and edge indicator e between them. Every selected G_H is positive definite. For row $i \ge 68$, take $H = H_i$ and $m = |H_i|$ in these definitions; thus h_σ, $z_{\sigma\tau}(e)$ and D belong to that row's reference graph. Define the row explicitly by

$$\begin{aligned}
R_i(y) = &-\sum_{\sigma} f_i(\sigma)^2 h_\sigma^{\ell_i} y_{(H_i)_\sigma}\\
&- \frac{404 - m}{2^{m+1}} \sum_{\sigma,\tau} \sum_{e=0}^{1} f_i(\sigma) f_i(\tau) \kappa_{\ell_i}(z_{\sigma\tau}(e), h_\sigma, h_\tau)\, y_{(H_i)_{\sigma,\tau;e}}.
\end{aligned} \tag{S1.1}$$

The convention $h^0 = 1$ is used. Isomorphic extensions are combined with their full multiplicities. Formula (S1.1) is the negative normalized kernel energy, so $R_i(y) \le 0$. For a fixed anchor, let $q_x \in \mathbb{R}^D$, $D = 30 - m$, be the orthogonal residuals of p_x for vertices outside the anchor.

With arbitrary real weights a_x, the sums of $a_x a_y \kappa_0$ and $a_x a_y \kappa_1$ are $(\sum_x a_x)^2$ and $\|\sum_x a_x q_x\|^2$, respectively, while

$$\sum_{x,y} a_x a_y \kappa_2(q_x^T q_y, \|q_x\|^2, \|q_y\|^2) = \frac{D}{D-1} \left\| \sum_x a_x \left(q_x q_x^T - \frac{\|q_x\|^2}{D} I_D \right) \right\|_F^2 \geq 0.$$

Here $\|\cdot\|_F$ is the Frobenius norm. Take $a_x = f_i(\sigma(x))$ and sum over all labelled embeddings of the anchor. Multiplication by the positive factor $2^{\binom{m+1}{2}}/(405)_{m+1}$ gives the negative of (S1.1). Hence $R_i(y) \leq 0$.

Here is a finite recursion specifying all ordinary counting substitutions. Let $\mathcal{C}$ contain one graph of each isomorphism type on at most seven vertices with minimum degree at least three. It has 173 elements. Introduce $x_C = y_C$ for $C \in \mathcal{C}$. Graph arguments are understood up to isomorphism: set $\mathcal{T}(K_1) = 1$ and $\mathcal{T}(F) = x_C$ whenever $F \cong C \in \mathcal{C}$. Fix a labelled representative for each type; for definiteness one may choose the representative with the least integer edge mask. For any other type F on at least two vertices, choose a vertex of minimum degree, delete it to obtain an m-vertex graph H, and let $B \subseteq [m]$ be its neighbour set. Then $|B| \leq 2$, and set

$$\mathcal{T}(F) = \frac{2^m e(H,B)}{405 - m} \mathcal{T}(H) - \sum_{B \subsetneq B' \subseteq [m]} \mathcal{T}(H_{B'}), \tag{S1.2}$$

where

$$e(H,B) = \begin{cases} 405 - m, & B = \varnothing, \\ 132 - \deg_H(b), & B = \{b\}, \\ 63 - |\mathcal{N}_H(b) \cap \mathcal{N}_H(c)|, & B = \{b,c\},\ b \sim_H c, \\ 33 - |\mathcal{N}_H(b) \cap \mathcal{N}_H(c)|, & B = \{b,c\},\ b \not\sim_H c. \end{cases}$$

Choose the least labelled vertex to break ties, and retain the increasing order of the surviving labels after deletion. The recursion terminates because each term has fewer vertices, or the same number of vertices and more edges. It specifies $\mathcal{T}(F)$ using only the ordinary one-vertex counting equations. The saved independent reconstruction checks all these equations and the identity rows for the free types. Induction through the recursion then shows that any choice of representatives gives the same affine expression in the isomorphism-type variables x_C.

Substitute $y_F = \mathcal{T}(F)$ in every printed source and collect terms:

$$\sum_{i=1}^{174} w_i R_i(\mathcal{T}) = c_0 + \sum_{C \in \mathcal{C}} c_C x_C. \tag{S1.3}$$

This equality defines the rational coefficients completely from the tables and formulas above. Put $U_C = 2^{\binom{|C|}{2}}$ and $E = -\sum_{c_C < 0} c_C U_C$. Exact substitution gives

$$c_0 > \frac{570}{10^{10}}, \qquad 0 \leq E < \frac{11}{10^{10}}. \tag{S1.4}$$

These are strict rational comparisons, not rounded decimal estimates. The original nonnegative counts and empty-set extension equations imply $0 \leq x_C \leq U_C$. Thus the fully specified certificate is

$$\begin{aligned} 0 \ &\geq \sum_{i=1}^{174} w_i R_i(\mathcal{T}) + \sum_{c_C \geq 0} c_C(-x_C) + \sum_{c_C < 0} (-c_C)(x_C - U_C) \\ &= c_0 - E > \frac{559}{10^{10}} > 0. \end{aligned} \tag{S1.5}$$

All 174 printed w_i are strictly positive. The direction arrays and rational coefficients below were generated from the exact inputs and checked against them entry by entry. In particular, the three second-degree rows are $i = 75, 76, 85$, all with $(m, b) = (5, 222)$, namely $K_{2,3}$ with bipartition $\{0,1\} \cup \{2,3,4\}$.

Table S1: All 67 count sources: $R_i = -y_{F_i}$. The two halves continue the same list.

i	$(\lvert F_i\rvert, b_i)$	w_i	i	$(\lvert F_i\rvert, b_i)$	w_i
1	$(6, 16320)$	$\frac{4432931675086999}{28800000000000000000}$	35	$(7, 2096133)$	$\frac{6747144395871761}{151040000000000000000}$
2	$(6, 32704)$	$\frac{293735452161148677}{15616000000000000000000}$	36	$(7, 236989)$	$\frac{286267699178147477}{800000000000000000000}$
3	$(6, 16322)$	$\frac{877203395721931}{1600000000000000000}$	37	$(7, 879100)$	$\frac{137770506922513731}{1856000000000000000000}$
4	$(6, 20286)$	$\frac{3451229927499611}{100000000000000000000}$	38	$(7, 1986296)$	$\frac{332436897606541119}{793600000000000000000}$
5	$(6, 20383)$	$\frac{15830843590921873}{100000000000000000000}$	39	$(7, 1589219)$	$\frac{1435021605037413981}{928000000000000000000}$
6	$(7, 31744)$	$\frac{12531988346210448997}{34967674880000000000000000}$	40	$(7, 1826416)$	$\frac{2123098113746581}{200000000000000000000}$
7	$(7, 135533)$	$\frac{1141661167868739 49}{16517038080000000000000}$	41	$(7, 1048069)$	$\frac{1668022362929567}{50000000000000000000}$
8	$(7, 16322)$	$\frac{842876737944493259}{19686400000000000000000}$	42	$(7, 1812083)$	$\frac{12466509026190 59}{40000000000000000000}$
9	$(7, 1070757)$	$\frac{748844430532683919}{32268800000000000000000}$	43	$(7, 1031904)$	$\frac{5326029766874909}{250000000000000000000}$
10	$(7, 54949)$	$\frac{14499381457879619}{1171200000000000000000}$	44	$(7, 2010852)$	$\frac{83787048882073}{2000000000000000000}$
11	$(7, 804336)$	$\frac{6972705278438623423}{58187776000000000000000}$	45	$(7, 522068)$	$\frac{5215829357476963}{100000000000000000000}$
12	$(7, 1333347)$	$\frac{41029753638653690845 83}{15361048576000000000000000}$	46	$(7, 980401)$	$\frac{18404445824673}{1000000000000000000}$
13	$(7, 706085)$	$\frac{1717793083371679901 01}{55700070400000000000000000}$	47	$(7, 1779319)$	$\frac{94173189168359}{2000000000000000000}$
14	$(7, 54957)$	$\frac{204178978336526613}{6400000000000000000000}$	48	$(7, 577503)$	$\frac{417198797138760903}{7552000000000000000000}$
15	$(7, 1510431)$	$\frac{795305301537721541}{9024000000000000000000}$	49	$(7, 1048480)$	$\frac{9350344806023369883}{1856000000000000000000000}$
16	$(7, 1048064)$	$\frac{728634566696438857}{5568000000000000000000}$	50	$(7, 122622)$	$\frac{84235295353712979}{2900000000000000000000}$
17	$(7, 220067)$	$\frac{5496089953577656587}{23168000000000000000000000}$	51	$(7, 2074227)$	$\frac{206632462572133}{6250000000000000000}$
18	$(7, 203768)$	$\frac{332857829316942321}{2432000000000000000000}$	52	$(7, 2079688)$	$\frac{698823698237899}{2000000000000000000000}$
19	$(7, 326195)$	$\frac{941500591424178322173}{48022323200000000000000000}$	53	$(7, 1899000)$	$\frac{1297200313453847}{2500000000000000000000}$
20	$(7, 1014793)$	$\frac{1095021535917172073}{31744000000000000000000}$	54	$(7, 912747)$	$\frac{10510941117098171}{25000000000000000000000}$
21	$(7, 1368429)$	$\frac{7397761749711252717}{49203200000000000000000000}$	55	$(7, 1556448)$	$\frac{4736859274093761}{2000000000000000000000}$
22	$(7, 1013836)$	$\frac{133381996368419 33}{31744000000000000000000}$	56	$(7, 1031664)$	$\frac{685234461081651}{1250000000000000000000}$
23	$(7, 2089528)$	$\frac{3518291171516883 23}{36249600000000000000000}$	57	$(7, 2039288)$	$\frac{3506682217891323}{1000000000000000000000}$
24	$(7, 101885)$	$\frac{4815145431912968 39}{15360000000000000000000}$	58	$(7, 515576)$	$\frac{706873785866791}{500000000000000000000}$
25	$(7, 103933)$	$\frac{1920298385396781 01}{16000000000000000000000}$	59	$(7, 1867768)$	$\frac{5432504025953709}{2500000000000000000000}$
26	$(7, 811517)$	$\frac{4253658482446455 81}{74240000000000000000000}$	60	$(7, 2080712)$	$\frac{1260473193245211}{2500000000000000000000}$
27	$(7, 1778239)$	$\frac{1998251094010137 53209}{15491072000000000000000000}$	61	$(7, 1048048)$	$\frac{3565258606328343}{1000000000000000000000}$
28	$(7, 980400)$	$\frac{1859742612814188781}{25600000000000000000000000}$	62	$(7, 2063243)$	$\frac{12162738640294731}{1000000000000000000000}$
29	$(7, 813933)$	$\frac{1296615683685705453}{64000000000000000000000000}$	63	$(7, 516088)$	$\frac{18512765334187303}{1000000000000000000000}$
30	$(7, 1032064)$	$\frac{95273191495385071}{16000000000000000000000}$	64	$(7, 1974265)$	$\frac{35433620623466983}{1000000000000000000000}$
31	$(7, 834040)$	$\frac{873631212813303}{2000000000000000000000}$	65	$(7, 1556427)$	$\frac{3736667720297609}{2000000000000000000000}$
32	$(7, 1693356)$	$\frac{279348814136549 37}{10000000000000000000000}$	66	$(7, 1048056)$	$\frac{127274694846441}{1000000000000000000}$
33	$(7, 946413)$	$\frac{255814985078879}{25000000000000000000}$	67	$(7, 2096632)$	$\frac{3837598246939311}{2000000000000000000000}$
34	$(7, 1779315)$	$\frac{2486850137261857}{12500000000000000000000}$			

Table S2: All 107 kernel sources. The source column gives $(m, b; \ell)$ and the final column gives w_i, followed by f_i row by row.

i	$(m, b; \ell)$	w_i and f_i
68	$(4, 45; 1)$	$w_i = \frac{35565071516699134399}{131890588161823486000000000000000000}$ $0\ -62492\ 152648\ -182005\ 62492\ 0\ -434322\ 463679$ $-152648\ 434322\ 0\ -189824\ 182005\ -463679\ 189824\ 0$
69	$(5, 621; 0)$	$w_i = \frac{8182262002905589}{7117363909909680000000000000000000}$ $0\ 0\ -102163\ 161206\ -36784\ 108851\ 266344\ -397454$ $36784\ -108851\ -2535\ 15557\ 0\ 0\ -161647\ 220691$ $102163\ -161206\ 0\ 0\ 2535\ -15557\ -232095\ 304161$ $-266344\ 397454\ 232095\ -304161\ 161647\ -220691\ 0\ 0$
70	$(5, 222; 1)$	$w_i = \frac{61597393293839903 67}{732888538006926353840000000000000000000}$ $0\ 347575\ -303289\ -123\ -44286\ -844\ 968\ 0$ $0\ -73921\ 64502\ -42033\ 9419\ -287861\ 329895\ 0$ $0\ -73921\ 64502\ -42033\ 9419\ -287861\ 329895\ 0$ $0\ -653\ 570\ 58824\ 83\ 402851\ -461675\ 0$
71	$(5, 478; 0)$	$w_i = \frac{2323334349629557}{16414562146500000000000000000000000000}$ $0\ -419980\ 0\ 228562\ 419980\ 0\ -228562\ 0$ $0\ 314220\ 0\ -122802\ -314220\ 0\ 122802\ 0$ $0\ 314220\ 0\ -122802\ -314220\ 0\ 122802\ 0$ $0\ -208460\ 0\ 17042\ 208460\ 0\ -17042\ 0$
72	$(5, 1021; 0)$	$w_i = \frac{3865479434076091}{12283033356272437500000000000000000000}$ $0\ 0\ 0\ 0\ -64660\ 175455\ 175455\ -286251$ $0\ 0\ 0\ 0\ 175455\ -286251\ -286251\ 397047$ $64660\ -175455\ -175455\ 286251\ 0\ 0\ 0\ 0$ $-175455\ 286251\ 286251\ -397047\ 0\ 0\ 0\ 0$
73	$(5, 1023; 1)$	$w_i = \frac{491278683697297}{27628860982380000000000000000000000000}$ $-190146\ 61106\ 61106\ 76994\ 61106\ 76994\ 76994\ -87626$ $61106\ 76994\ 76994\ -87626\ 76994\ -87626\ -87626\ -165735$ $61106\ 76994\ 76994\ -87626\ 76994\ -87626\ -87626\ -165735$ $76994\ -87626\ -87626\ -165735\ -87626\ -165735\ -165735\ 819616$
74	$(5, 960; 0)$	$w_i = \frac{89599552921041407}{13015616157077691164000000000000000000}$ $0\ 0\ -40858\ -174319\ 7002\ 29873\ 16920\ 161383$ $-48166\ -205497\ 44491\ 424349\ 20572\ 196215\ 38\ -432004$ $82023\ 349943\ -20572\ -196215\ -44491\ -424349\ -23\ 253685$ $-16920\ -161383\ 3\ -36878\ -19\ 215197\ 0\ 0$
75	$(5, 222; 2)$	$w_i = \frac{916955097401716893}{50549692936109292447500000000000000000}$ $0\ 619508\ -369656\ 13124\ -249851\ 19417\ -32540\ 0$ $0\ -330870\ 197428\ 49736\ 133442\ 73585\ -123321\ 0$ $0\ -330870\ 197428\ 49736\ 133442\ 73585\ -123321\ 0$ $0\ -25549\ 15245\ 61946\ 10304\ 91649\ -153595\ 0$
76	$(5, 222; 2)$	$w_i = \frac{386746425034437}{53675406395459206542968750000000}$ $117610\ -293584\ -293584\ -14274\ -293584\ -14274\ -14274\ 0$ $-34929\ -18491\ -18491\ 265514\ -18491\ 265514\ 265514\ 52070$ $-34929\ -18491\ -18491\ 265514\ -18491\ 265514\ 265514\ 52070$ $0\ -13820\ -13820\ -290273\ -13820\ -290273\ -290273\ 201821$
77	$(5, 478; 0)$	$w_i = \frac{7393629964078759}{11028270858412500000000000000000000000}$ $0\ -5318\ 0\ -231999\ 5318\ 0\ 231999\ 0$ $0\ -90660\ 0\ 327977\ 90660\ 0\ -327977\ 0$ $0\ -90660\ 0\ 327977\ 90660\ 0\ -327977\ 0$ $0\ 186639\ 0\ -423955\ -186639\ 0\ 423955\ 0$
78	$(5, 1005; 1)$	$w_i = \frac{716954813220737}{12297880462118565000000000000000000000}$ $64187\ 433525\ 59071\ -208362\ 59071\ -208362\ -199932\ 67356$ $59071\ -208362\ -30469\ -62510\ -199932\ 67356\ 124176\ 184115$ $59071\ -208362\ -199932\ 67356\ -30469\ -62510\ 124176\ 184115$ $-199932\ 67356\ 124176\ 184115\ 124176\ 184115\ 63490\ -480943$

i	$(m,b;\ell)$	w_i and f_i
79	$(5,1021;0)$	$w_i = \frac{2083331004868951}{10320484797707719680000000000000000000}$ $\begin{matrix} -399921 & 388647 & 388647 & -86977 & 369889 & -41016 & -41016 & 19719 \\ 388647 & -86977 & -86977 & -916 & -41016 & 19719 & 19719 & -15807 \\ 369889 & -41016 & -41016 & 19719 & 4194 & -52694 & -52694 & -7058 \\ -41016 & 19719 & 19719 & -15807 & -52694 & -7058 & -7058 & 262999 \end{matrix}$
80	$(5,1023;1)$	$w_i = \frac{207266842177493}{14928940192996667187500000000000000}$ $\begin{matrix} 0 & 31426 & 61125 & -229740 & 6231 & -93477 & -167198 & 231681 \\ -143942 & 279300 & 205579 & -120533 & 341842 & -249279 & -179624 & 26609 \\ 45160 & -190111 & -263833 & 322985 & -127570 & 194239 & 263894 & -84813 \\ 245207 & -157975 & -88320 & 3671 & -217067 & 36016 & 18517 & 0 \end{matrix}$
81	$(5,992;1)$	$w_i = \frac{2397606624605478069}{10761518622544307989760000000000000000}$ $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 140279 & 289318 & -83842 & -307654 & -83842 & -307654 & -79047 & 432443 \\ -140279 & -289318 & 83842 & 307654 & 83842 & 307654 & 79047 & -432443 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
82	$(5,729;1)$	$w_i = \frac{21910999623230153}{39733433441334633400000000000000000}$ $\begin{matrix} 52995 & -126974 & -126974 & 233401 & 193930 & -204270 & -204270 & 98115 \\ -126974 & 233401 & 233401 & -166734 & -204270 & 98115 & 98115 & -81008 \\ -126974 & 233401 & 233401 & -166734 & -204270 & 98115 & 98115 & -81008 \\ 233401 & -166734 & -166734 & -278573 & 98115 & -81008 & -81008 & 358494 \end{matrix}$
83	$(5,437;0)$	$w_i = \frac{2791812622047061657}{31607950223891414796800000000000000000}$ $\begin{matrix} -117793 & 101558 & 101558 & 99305 & -97248 & 15131 & 15131 & -248244 \\ -80543 & 80178 & 80178 & -374556 & -118473 & 107714 & 107714 & 156174 \\ 160685 & -238564 & -238564 & 1214 & -180995 & 226624 & 226624 & -558 \\ 33009 & -68491 & -68491 & 355822 & 291882 & -287530 & -287530 & -9004 \end{matrix}$
84	$(5,222;1)$	$w_i = \frac{4877206172843601}{46850227268723200000000000000000000}$ $\begin{matrix} 0 & 121110 & -369946 & 119 & 248836 & -178 & 58 & 0 \\ 0 & 89238 & -272589 & 139162 & 183351 & -206893 & 67731 & 0 \\ 0 & 89238 & -272589 & 139162 & 183351 & -206893 & 67731 & 0 \\ 0 & 5971 & -18240 & 349364 & 12269 & -519400 & 170037 & 0 \end{matrix}$
85	$(5,222;2)$	$w_i = \frac{935932178379011067}{20528165536230077984600000000000000000}$ $\begin{matrix} 0 & -490381 & 506130 & 585 & -15749 & -18788 & 18203 & 0 \\ 0 & 151976 & -156857 & -10059 & 4881 & 323273 & -313214 & 0 \\ 0 & 151976 & -156857 & -10059 & 4881 & 323273 & -313214 & 0 \\ 0 & 13296 & -13723 & -771 & 427 & 24784 & -24013 & 0 \end{matrix}$
86	$(5,1017;1)$	$w_i = \frac{6051359310862721}{57013646012951013600000000000000000000}$ $\begin{matrix} 0 & -328898 & 328898 & 0 & 0 & 9247 & -9247 & 0 \\ 0 & 382723 & -382723 & 0 & 0 & -157371 & 157371 & 0 \\ 0 & 382723 & -382723 & 0 & 0 & -157371 & 157371 & 0 \\ 0 & -208124 & 208124 & 0 & 0 & 77072 & -77072 & 0 \end{matrix}$
87	$(5,1005;0)$	$w_i = \frac{1091886607993667}{79953173520691200000000000000000000000}$ $\begin{matrix} -95956 & 50765 & 238421 & -153404 & 238421 & -153404 & -263515 & 273705 \\ 238421 & -153404 & -20190 & 9199 & -263515 & 273705 & 62945 & -102812 \\ 238421 & -153404 & -263515 & 273705 & -20190 & 9199 & 62945 & -102812 \\ -263515 & 273705 & 62945 & -102812 & 62945 & -102812 & 164973 & -54391 \end{matrix}$
88	$(5,1005;0)$	$w_i = \frac{1654874217157843}{62343612249738750000000000000000000}$ $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 353553 & -353553 \\ 0 & 0 & 0 & 0 & -353553 & 353553 & 0 & 0 \\ 0 & 0 & -353553 & 353553 & 0 & 0 & 0 & 0 \\ 353553 & -353553 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
89	$(5,1005;1)$	$w_i = \frac{887493888104349}{37881487439190440000000000000000000}$ $\begin{matrix} 276193 & -175335 & -200474 & 220557 & -200474 & 220557 & -77445 & -132673 \\ -200474 & 220557 & 10665 & 18916 & -77445 & -132673 & 150399 & 79151 \\ -200474 & 220557 & -77445 & -132673 & 10665 & 18916 & 150399 & 79151 \\ -77445 & -132673 & 150399 & 79151 & 150399 & 79151 & 212558 & -530634 \end{matrix}$

i	$(m,b;\ell)$	w_i and f_i
90	$(5,1005;1)$	$w_i = \frac{5120966254311011}{72858393691420312500000000000000000}$ $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & -283979 & 411529 \\ 0 & 0 & 0 & 0 & 283979 & -411529 & 0 & 0 \\ 0 & 0 & 283979 & -411529 & 0 & 0 & 0 & 0 \\ -283979 & 411529 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
91	$(5,1021;0)$	$w_i = \frac{1846732871507397}{59082311397190400000000000000000000}$ $\begin{matrix} 110 & -87821 & -87821 & 251457 & 134492 & -45120 & -45120 & -65154 \\ -87821 & 251457 & 251457 & -444627 & -45120 & -65154 & -65154 & 223033 \\ 134492 & -45120 & -45120 & -65154 & 136105 & -172115 & -172115 & 247369 \\ -45120 & -65154 & -65154 & 223033 & -172115 & 247369 & 247369 & -353775 \end{matrix}$
92	$(5,1023;1)$	$w_i = \frac{18518492245803023}{52346964604387218750000000000000000}$ $\begin{matrix} 38454 & 229458 & 229458 & -110605 & 229458 & -110605 & -110605 & -144064 \\ 229458 & -110605 & -110605 & -144064 & -110605 & -144064 & -144064 & 283596 \\ 229458 & -110605 & -110605 & -144064 & -110605 & -144064 & -144064 & 283596 \\ -110605 & -144064 & -144064 & 283596 & -144064 & 283596 & 283596 & -57042 \end{matrix}$
93	$(4,52;0)$	$w_i = \frac{3532848272568855501}{940832706473123648000000000000000000}$ $\begin{matrix} 17908 & -45541 & -45541 & 173550 & -45541 & 173550 & 173550 & -244952 \\ -125267 & 271367 & 271367 & -359693 & 271367 & -359693 & -359693 & 467277 \end{matrix}$
94	$(4,45;1)$	$w_i = \frac{8926077345717279209}{83937068175660299200000000000000000000}$ $\begin{matrix} -175075 & 151903 & 151903 & 44339 & 151903 & -75637 & 44339 & -293674 \\ 151903 & 44339 & -75637 & -293674 & 44339 & -293674 & -293674 & 716078 \end{matrix}$
95	$(4,47;1)$	$w_i = \frac{3771971858146137}{14789383580268155200000000000000000}$ $\begin{matrix} 32547 & 5720 & 256576 & -90314 & 5720 & -334335 & -90314 & 214400 \\ 256576 & -90314 & -654571 & 283780 & -90314 & 214400 & 283780 & -203337 \end{matrix}$
96	$(5,5;0)$	$w_i = \frac{9517576461589525093}{120358330404505204736000000000000000000000}$ $\begin{matrix} 0 & 0 & 0 & 0 & 22061 & -237066 & -237066 & 452070 \\ 0 & 0 & 0 & 0 & 64761 & 150243 & 150243 & -365248 \\ -22061 & 237066 & 237066 & -452070 & 0 & 0 & 0 & 0 \\ -64761 & -150243 & -150243 & 365248 & 0 & 0 & 0 & 0 \end{matrix}$
97	$(5,517;0)$	$w_i = \frac{10438665066265197}{24540654135258729920000000000000000000}$ $\begin{matrix} 0 & 0 & 0 & 0 & 21098 & -158515 & -158515 & 295933 \\ 0 & 0 & 0 & 0 & -147169 & 284586 & 284586 & -422004 \\ -21098 & 158515 & 158515 & -295933 & 0 & 0 & 0 & 0 \\ 147169 & -284586 & -284586 & 422004 & 0 & 0 & 0 & 0 \end{matrix}$
98	$(5,960;1)$	$w_i = \frac{107394004274549959}{30010951631819714950000000000000000}$ $\begin{matrix} 0 & 0 & -4366 & 30550 & -35732 & 250006 & -139430 & -253713 \\ -552 & 3863 & -17103 & -31121 & -126167 & -229579 & 529 & 552815 \\ 40650 & -284420 & 126167 & 229579 & 17103 & 31121 & -7 & -7509 \\ 139430 & 253713 & -465 & -485927 & -57 & -59379 & 0 & 0 \end{matrix}$
99	$(5,101;0)$	$w_i = \frac{1555749759454044 59}{18303284905432282434560000000000000000}$ $\begin{matrix} 0 & 142602 & -142602 & 0 & 45094 & -199320 & 260948 & -106723 \\ 0 & -151729 & 151729 & 0 & -55436 & 218788 & -259734 & 96381 \\ -45094 & -260948 & 199320 & 106723 & 0 & 317666 & -317666 & 0 \\ 55436 & 259734 & -218788 & -96381 & 0 & -326793 & 326793 & 0 \end{matrix}$
100	$(5,519;0)$	$w_i = \frac{441877492854601393}{89600312701096358016000000000000000000}$ $\begin{matrix} -31443 & -27260 & -27260 & 245501 & -55433 & 29586 & 29586 & -328447 \\ -55433 & 29586 & 29586 & -328447 & -108997 & 78755 & 78755 & 220594 \\ -55433 & 29586 & 29586 & -328447 & -108997 & 78755 & 78755 & 220594 \\ -108997 & 78755 & 78755 & 220594 & 360381 & -353364 & -353364 & 26544 \end{matrix}$
101	$(5,992;0)$	$w_i = \frac{1983081816301178287}{48461281246051741922560000000000000000}$ $\begin{matrix} -57919 & 313867 & 170247 & -345639 & 170247 & -345639 & -192064 & 261387 \\ -11955 & -178429 & -136151 & 220316 & -136151 & 220316 & 163133 & -155936 \\ -11955 & -178429 & -136151 & 220316 & -136151 & 220316 & 163133 & -155936 \\ 193873 & -99369 & -40603 & 37461 & -40603 & 37461 & -1747 & -91275 \end{matrix}$

i	$(m,b;\ell)$	w_i and f_i
102	$(5,992;0)$	$w_i = \frac{1740690333499091}{26738626765115520000000000000000000}$
		$\begin{matrix} 0 & 0 & -52475 & -196996 & 52475 & 196996 & 0 & 0 \\ 0 & 0 & -72942 & 322413 & 72942 & -322413 & 0 & 0 \\ 0 & 0 & -72942 & 322413 & 72942 & -322413 & 0 & 0 \\ 0 & 0 & 198359 & -447830 & -198359 & 447830 & 0 & 0 \end{matrix}$
103	$(5,992;1)$	$w_i = \frac{8940520736711881}{432770961428472799100000000000000000}$
		$\begin{matrix} -59517 & -88101 & -100287 & 356895 & -100287 & 356895 & 81770 & -448950 \\ 14723 & -247818 & -56656 & 73412 & -56656 & 73412 & -6200 & 207364 \\ 14723 & -247818 & -56656 & 73412 & -56656 & 73412 & -6200 & 207364 \\ 361455 & 159928 & 13311 & -211006 & 13311 & -211006 & -181 & -127394 \end{matrix}$
104	$(5,103;0)$	$w_i = \frac{2477826471212474797}{25136580219485907392000000000000000000}$
		$\begin{matrix} -3080 & -128153 & -128153 & 473296 & -1881 & -45553 & 63975 & -207466 \\ 95835 & 21791 & 21791 & -330230 & -101728 & 182243 & 76225 & 41770 \\ -1881 & 63975 & -45553 & -207466 & -39545 & 175480 & 175480 & -112904 \\ -101728 & 76225 & 182243 & 41770 & 173368 & -334912 & -334912 & 304964 \end{matrix}$
105	$(5,433;1)$	$w_i = \frac{3524220686707390819}{73312484349944163951200000000000000000}$
		$\begin{matrix} 62964 & -93023 & -93023 & 3504 & 163544 & 82507 & 82507 & -505954 \\ -295911 & 36470 & 36470 & 77465 & 378 & 124034 & 124034 & 194035 \\ -198643 & 236437 & 236437 & 522 & -259028 & 144047 & 144047 & -6844 \\ 319689 & -221400 & -221400 & 113442 & 41553 & -143616 & -143616 & -41625 \end{matrix}$
106	$(5,433;1)$	$w_i = \frac{7504571763760729033}{36510001417106580349640000000000000000}$
		$\begin{matrix} 1226 & -11376 & -11376 & -43256 & -178546 & 99820 & 99820 & -210643 \\ 368046 & -234573 & -234573 & 467838 & 3841 & -36509 & -36509 & -43231 \\ -24157 & 115482 & 115482 & 426 & -156928 & 94479 & 94479 & 15067 \\ -194633 & -92280 & -92280 & -129999 & -65893 & 312000 & 312000 & -303247 \end{matrix}$
107	$(5,613;0)$	$w_i = \frac{8592577785919841}{21856521898888953600000000000000000}$
		$\begin{matrix} 0 & 19920 & 44302 & -234153 & 7460 & -34066 & -188722 & 385259 \\ -39691 & 24599 & -5736 & 190759 & 117516 & -95738 & 64872 & -256581 \\ -31990 & 44008 & -15318 & 173231 & 30521 & -35852 & 153748 & -318347 \\ 261351 & -278196 & -212917 & 59832 & -345166 & 355325 & 159771 & 0 \end{matrix}$
108	$(5,437;0)$	$w_i = \frac{1268231699898718381}{16935099701554176273920000000000000000}$
		$\begin{matrix} 24208 & 61785 & 61785 & 92572 & -228891 & -68416 & -68416 & 188323 \\ -233721 & 203876 & 203876 & -361305 & 365353 & -71427 & -71427 & -15169 \\ -1849 & 50513 & 50513 & -276575 & 109634 & 115967 & 115967 & -143186 \\ 153802 & -195970 & -195970 & 446088 & -135768 & -65779 & -65779 & 78818 \end{matrix}$
109	$(5,945;1)$	$w_i = \frac{60127745932404464559}{12856883126300000341408000000000000000000}$
		$\begin{matrix} 16780 & -131392 & -131392 & 179364 & 137525 & -56269 & -56269 & -322219 \\ -146402 & 260535 & 260535 & 62895 & 6687 & -37122 & -37122 & -6131 \\ -57381 & 48312 & 48312 & 83450 & -275158 & 217412 & 217412 & 81517 \\ 309433 & -173679 & -173679 & -455688 & -96056 & -23222 & -23222 & 272238 \end{matrix}$
110	$(5,993;0)$	$w_i = \frac{8918917090767866327}{18100344064088401856000000000000000000}$
		$\begin{matrix} 116821 & -70907 & -132646 & 215205 & -132646 & 215205 & 46004 & -61693 \\ -132646 & 215205 & 168951 & -213651 & 168951 & -213651 & 94488 & -21888 \\ -132646 & 215205 & 168951 & -213651 & 168951 & -213651 & 94488 & -21888 \\ 46004 & -61693 & 94488 & -21888 & 94488 & -21888 & -468965 & 399409 \end{matrix}$
111	$(5,993;1)$	$w_i = \frac{1971639743210093}{10513113337945951200000000000000000}$
		$\begin{matrix} 0 & 0 & 77668 & -292495 & 77668 & -292495 & 161223 & 318355 \\ -77668 & 292495 & 0 & 0 & 0 & 0 & 40390 & -305141 \\ -77668 & 292495 & 0 & 0 & 0 & 0 & 40390 & -305141 \\ -161223 & -318355 & -40390 & 305141 & -40390 & 305141 & 0 & 0 \end{matrix}$
112	$(5,621;1)$	$w_i = \frac{293886947787454393}{10854269109149307321000000000000000000}$
		$\begin{matrix} 39978 & -157957 & 15395 & -52865 & -108767 & 279770 & 114505 & -293373 \\ -108767 & 279770 & -88976 & -29685 & -405779 & 182958 & 227149 & 106642 \\ 15395 & -52865 & -41200 & 335100 & -88976 & -29685 & 35719 & -10175 \\ 114505 & -293373 & 35719 & -10175 & 227149 & 106642 & 16951 & -360731 \end{matrix}$

i	$(m,b;\ell)$	w_i and f_i
113	$(5,222;1)$	$w_i = \dfrac{28258627022212189}{15207733184360847158000000000000000}$
		$\begin{matrix} 129598 & 302064 & 302064 & -1164 & 302064 & -1164 & -1164 & 0 \\ 28737 & -191406 & -191406 & -159040 & -191406 & -159040 & -159040 & -9697 \\ 28737 & -191406 & -191406 & -159040 & -191406 & -159040 & -159040 & -9697 \\ 0 & -17877 & -17877 & 329422 & -17877 & 329422 & 329422 & 97662 \end{matrix}$
114	$(5,1012;0)$	$w_i = \dfrac{15529849714389967}{20158207303631200000000000000000000}$
		$\begin{matrix} 0 & 0 & 19043 & -9869 & -86875 & 45021 & 193935 & -161256 \\ 67832 & -35152 & -248381 & 206527 & 54446 & -45272 & 0 & 0 \\ 0 & 0 & -61342 & 52167 & 279840 & -237986 & -344601 & 311922 \\ -218498 & 185818 & 441346 & -399492 & -96745 & 87570 & 0 & 0 \end{matrix}$
115	$(5,1016;0)$	$w_i = \dfrac{1334751114552388891}{11285211164350150200000000000000000000}$
		$\begin{matrix} -28548 & -194453 & -194453 & 45361 & -60973 & 238702 & 238702 & 27031 \\ -60973 & 238702 & 238702 & 27031 & 99987 & -147302 & -147302 & -207526 \\ -60973 & 238702 & 238702 & 27031 & 99987 & -147302 & -147302 & -207526 \\ 99987 & -147302 & -147302 & -207526 & -3352 & -51366 & -51366 & 521167 \end{matrix}$
116	$(5,1016;0)$	$w_i = \dfrac{205465028551129}{89354296356822500000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & -8049 & -37157 & -37157 & 82363 \\ 2359 & 10889 & 10889 & -24136 & -245685 & 277643 & 277643 & -309602 \\ 5690 & 26268 & 26268 & -58227 & -101842 & 115089 & 115089 & -128337 \\ 347527 & -392733 & -392733 & 437939 & 0 & 0 & 0 & 0 \end{matrix}$
117	$(5,1016;1)$	$w_i = \dfrac{2374238923720170331}{53779605798296070240000000000000000000}$
		$\begin{matrix} -39498 & 20340 & 20340 & -57339 & 224897 & -55205 & -55205 & 140726 \\ 224897 & -55205 & -55205 & 140726 & -161048 & -643 & -643 & -291937 \\ 224897 & -55205 & -55205 & 140726 & -161048 & -643 & -643 & -291937 \\ -161048 & -643 & -643 & -291937 & -6246 & 1400 & 1400 & 656776 \end{matrix}$
118	$(5,757;0)$	$w_i = \dfrac{7778090159809309}{11439515301130582400000000000000000000}$
		$\begin{matrix} 9397 & 50992 & 50992 & -80910 & -159706 & 40031 & 161617 & 62661 \\ 231234 & -128820 & -128820 & 109821 & -115466 & 186224 & 85164 & -195636 \\ -159706 & 161617 & 40031 & 62661 & 541342 & -348893 & -348893 & 96866 \\ -115466 & 85164 & 186224 & -195636 & -96554 & -1954 & -1954 & 181227 \end{matrix}$
119	$(5,478;0)$	$w_i = \dfrac{14506664330807267}{18089855369164800000000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2844 & 190362 & 43998 & -231516 & 190362 & -377880 & -231516 & 419033 \\ 2844 & -190362 & -43998 & 231516 & -190362 & 377880 & 231516 & -419033 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
120	$(5,478;1)$	$w_i = \dfrac{11607564714560783}{89722748462072000000000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 188578 & -70012 & -226592 & 354248 & -70012 & -341439 & 354248 & -189017 \\ -188578 & 70012 & 226592 & -354248 & 70012 & 341439 & -354248 & 189017 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
121	$(5,1017;1)$	$w_i = \dfrac{3057017586128491}{35694847171703928000000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -13973 & 150748 & 150748 & -281533 & 434532 & -289196 & -289196 & 137870 \\ 13973 & -150748 & -150748 & 281533 & -434532 & 289196 & 289196 & -137870 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
122	$(5,1005;1)$	$w_i = \dfrac{12321983632492903}{63992230848294656250000000000000000}$
		$\begin{matrix} 0 & 0 & 210787 & -176831 & 118565 & -99465 & -246005 & -191108 \\ -210787 & 176831 & 0 & 0 & 68884 & 53512 & 30812 & 264803 \\ -118565 & 99465 & -68884 & -53512 & 0 & 0 & 54779 & 470774 \\ 246005 & 191108 & -30812 & -264803 & -54779 & -470774 & 0 & 0 \end{matrix}$
123	$(5,1021;0)$	$w_i = \dfrac{51851331871681}{95879438899725000000000000000000000}$
		$\begin{matrix} -215410 & 101608 & 101608 & 162862 & 293446 & 15811 & 15811 & -117417 \\ 101608 & 162862 & 162862 & -275889 & 15811 & -117417 & -117417 & 185262 \\ 293446 & 15811 & 15811 & -117417 & 79448 & -225841 & -225841 & 284023 \\ 15811 & -117417 & -117417 & 185262 & -225841 & 284023 & 284023 & -201300 \end{matrix}$

i	$(m, b; \ell)$	w_i and f_i
124	$(5, 1021; 1)$	$w_i = \frac{176763181912949}{14537042980692750000000000000000}$
		$\begin{matrix} 112526 & 222404 & 222404 & -230419 & -22254 & -87036 & -87036 & 41490 \\ 222404 & -230419 & -230419 & -345719 & -87036 & 41490 & 41490 & 416129 \\ -22254 & -87036 & -87036 & 41490 & 53130 & -78832 & -78832 & 87291 \\ -87036 & 41490 & 41490 & 416129 & -78832 & 87291 & 87291 & -335743 \end{matrix}$
125	$(5, 1023; 0)$	$w_i = \frac{1463593012977857}{28758610014776250000000000000000}$
		$\begin{matrix} 0 & 4742 & 135592 & -196785 & -17938 & 18504 & -164981 & 220865 \\ -161723 & 220127 & 36642 & -38596 & 251931 & -315644 & -79523 & 86785 \\ 39327 & -61796 & -245281 & 324201 & -29992 & 47153 & 283274 & -356885 \\ 171631 & -212308 & 23812 & -39585 & -253235 & 299221 & 10465 & 0 \end{matrix}$
126	$(4, 56; 0)$	$w_i = \frac{1624697887902654849}{20499523309862295680000000000000000}$
		$\begin{matrix} 17548 & 132437 & -26270 & 103227 & -26270 & 103227 & 305731 & -251300 \\ -26270 & 103227 & 305731 & -251300 & 305731 & -251300 & -494946 & 482984 \end{matrix}$
127	$(4, 56; 1)$	$w_i = \frac{2237150249890325499}{18827903521042665740000000000000000}$
		$\begin{matrix} -225329 & 539866 & -31685 & -278873 & -31685 & -278873 & 57374 & 249207 \\ -31685 & -278873 & 57374 & 249207 & 57374 & 249207 & 148263 & -450866 \end{matrix}$
128	$(4, 60; 0)$	$w_i = \frac{254018830525342133}{30027845200197838800000000000000000}$
		$\begin{matrix} -53968 & -30569 & 110699 & -112274 & 110699 & -112274 & -269563 & 222075 \\ 164251 & -181347 & -323116 & 290647 & -323116 & 290647 & 447936 & -431488 \end{matrix}$
129	$(4, 60; 1)$	$w_i = \frac{14094485914483465883}{88901221780586464080000000000000000}$
		$\begin{matrix} -68565 & 355954 & 68934 & -412950 & 68934 & -412950 & 68421 & 332223 \\ -140125 & -218763 & 98344 & 317172 & 98344 & 317172 & -194287 & -277857 \end{matrix}$
130	$(4, 63; 1)$	$w_i = \frac{78185258104877}{36219703675764450000000000000000000}$
		$\begin{matrix} 91584 & 160136 & 160136 & -289633 & 160136 & -289633 & -289633 & 296904 \\ 160136 & -289633 & -289633 & 296904 & -289633 & 296904 & 296904 & -181952 \end{matrix}$
131	$(5, 5; 0)$	$w_i = \frac{69885431881186891}{37742141529284049510400000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 96618 & 123739 & 123739 & -344096 \\ 0 & 0 & 0 & 0 & 26892 & -247248 & -247248 & 467605 \\ -96618 & -123739 & -123739 & 344096 & 0 & 0 & 0 & 0 \\ -26892 & 247248 & 247248 & -467605 & 0 & 0 & 0 & 0 \end{matrix}$
132	$(5, 560; 0)$	$w_i = \frac{729552130954857259}{31345323979681870550400000000000000000}$
		$\begin{matrix} 77460 & 54998 & 87414 & -25160 & 54998 & -48678 & -25160 & -166211 \\ 54998 & -48678 & -25160 & -166211 & -48678 & -86645 & -166211 & 492154 \\ 38975 & -154337 & -373752 & 304063 & -154337 & 142458 & 304063 & -101507 \\ -154337 & 142458 & 304063 & -101507 & 142458 & 2289 & -101507 & -232041 \end{matrix}$
133	$(5, 560; 1)$	$w_i = \frac{58263927244992003}{18274538570296071427558400000000000000}$
		$\begin{matrix} 128124 & 3636 & -400895 & 147951 & 3636 & -45412 & 147951 & -149775 \\ 3636 & -45412 & 147951 & -149775 & -45412 & -51311 & -149775 & 454883 \\ 107139 & 76005 & -273114 & 245804 & 76005 & -94885 & 245804 & -217974 \\ 76005 & -94885 & 245804 & -217974 & -94885 & -1984 & -217974 & 141109 \end{matrix}$
134	$(5, 608; 0)$	$w_i = \frac{853536994858899983}{21712716326995963168000000000000000000}$
		$\begin{matrix} -32183 & -84477 & -84477 & -31893 & -8849 & -85315 & 85069 & 31910 \\ 125186 & -201979 & -201979 & 300350 & -131149 & 207315 & -1403 & -218762 \\ -8849 & 85069 & -85315 & 31910 & 214298 & -291144 & -291144 & 223993 \\ -131149 & -1403 & 207315 & -218762 & -278325 & 289084 & 289084 & -196651 \end{matrix}$
135	$(5, 608; 0)$	$w_i = \frac{3149021250202078981}{50041840870321066028800000000000000000}$
		$\begin{matrix} 2923 & 75245 & 75245 & -86547 & 130505 & -185649 & -177729 & 150533 \\ -20941 & -5383 & -5383 & -50724 & 96049 & -108315 & -119057 & 204742 \\ 130505 & -177729 & -185649 & 150533 & -180833 & 189467 & 189467 & -124681 \\ 96049 & -119057 & -108315 & 204742 & -272738 & 325323 & 325323 & -454332 \end{matrix}$
136	$(5, 608; 0)$	$w_i = \frac{7212844717445993733}{74566072330964211353600000000000000000}$
		$\begin{matrix} 0 & -130316 & 130316 & 0 & 5925 & 62867 & 100177 & -168968 \\ 0 & 87181 & -87181 & 0 & 202983 & -228639 & -352219 & 377875 \\ -5925 & -100177 & -62867 & 168968 & 0 & 167626 & -167626 & 0 \\ -202983 & 352219 & 228639 & -377875 & 0 & -210761 & 210761 & 0 \end{matrix}$

i	$(m,b;\ell)$	w_i and f_i
137	$(5,564;0)$	$w_i = \dfrac{141693271291691529}{52434231175425659200000000000000000000}$
		$\begin{matrix} -35413 & -76345 & 35986 & 102371 & -3791 & 163798 & -34649 & -127913 \\ -3791 & 163798 & -34649 & -127913 & -80666 & -103405 & 180876 & 19389 \\ 12446 & 303402 & -87020 & -227024 & -102600 & -232953 & 239085 & 108585 \\ -102600 & -232953 & 239085 & 108585 & 344959 & 24220 & -529434 & 152779 \end{matrix}$
138	$(5,45;0)$	$w_i = \dfrac{1492370246080549}{18529852619447504000000000000000000}$
		$\begin{matrix} 0 & 0 & 32208 & 103835 & -43423 & -139994 & -17680 & 65055 \\ -32208 & -103835 & 0 & 0 & -119220 & 438680 & 180323 & -363741 \\ 43423 & 139994 & 119220 & -438680 & 0 & 0 & -133748 & 269790 \\ 17680 & -65055 & -180323 & 363741 & 133748 & -269790 & 0 & 0 \end{matrix}$
139	$(5,960;1)$	$w_i = \dfrac{1201199563946272157}{18390634665380893511200000000000000000}$
		$\begin{matrix} 0 & 0 & 197348 & -378322 & 54803 & -105058 & -143369 & 294651 \\ -283216 & 542933 & 48823 & -100341 & 129872 & -266912 & 643 & 8146 \\ 31065 & -59552 & -129872 & 266912 & -48823 & 100341 & -5858 & -74264 \\ 143369 & -294651 & 1133 & 14370 & 4082 & 51748 & 0 & 0 \end{matrix}$
140	$(5,624;0)$	$w_i = \dfrac{1537666233516861473}{10361552426581658493440000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & -161367 & 260811 & 92769 & -192212 \\ 161367 & -260811 & -92769 & 192212 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 278598 & -378041 & -210000 & 309443 \\ -278598 & 378041 & 210000 & -309443 & 0 & 0 & 0 & 0 \end{matrix}$
141	$(5,101;0)$	$w_i = \dfrac{814213404213043221}{11378914750592589683200000000000000000}$
		$\begin{matrix} -36702 & -78538 & -78538 & 397289 & 114702 & 55717 & 68545 & -280662 \\ 2510 & 73468 & 73468 & -215853 & -142524 & 148470 & 159555 & -45687 \\ 114702 & 68545 & 55717 & -280662 & -75843 & -26278 & -26278 & 223502 \\ -142524 & 159555 & 148470 & -45687 & 301984 & -322822 & -322822 & 274065 \end{matrix}$
142	$(5,101;0)$	$w_i = \dfrac{84220113373528556 23}{14310956525563713817600000000000000000}$
		$\begin{matrix} 0 & 48664 & -48664 & 0 & 78336 & 17490 & 192751 & -288577 \\ 0 & -55034 & 55034 & 0 & 66063 & -155519 & -343519 & 432976 \\ -78336 & -192751 & -17490 & 288577 & 0 & 126596 & -126596 & 0 \\ -66063 & 343519 & 155519 & -432976 & 0 & -132966 & 132966 & 0 \end{matrix}$
143	$(5,101;0)$	$w_i = \dfrac{723189644764078673}{22296023893692439667200000000000000000}$
		$\begin{matrix} 24991 & -71345 & -71345 & 61445 & -156179 & 149592 & 17988 & 159948 \\ -92137 & 245387 & 245387 & -244866 & 109630 & -112420 & 36193 & -149188 \\ -156179 & 17988 & 149592 & 159948 & 437441 & -149356 & -149356 & -272089 \\ 109630 & 36193 & -112420 & -149188 & -180244 & -60476 & -60476 & 452700 \end{matrix}$
144	$(5,519;0)$	$w_i = \dfrac{2555709449439763}{30333438191365922800000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 15385 & -146919 & -146919 & 278453 \\ -14535 & 138795 & 138795 & -263056 & -9590 & 16863 & 16863 & -24137 \\ -851 & 8124 & 8124 & -15397 & -163840 & 288101 & 288101 & -412362 \\ 173430 & -304964 & -304964 & 436498 & 0 & 0 & 0 & 0 \end{matrix}$
145	$(5,47;1)$	$w_i = \dfrac{11588690942183502527}{62277293629935723439200000000000000000}$
		$\begin{matrix} 93376 & 104890 & -114456 & -91905 & 150679 & 41735 & -21983 & -127884 \\ -114456 & -91905 & -95689 & 35174 & -21983 & -127884 & 203906 & 178385 \\ 150679 & 41735 & -21983 & -127884 & -382737 & -177808 & 172193 & 311351 \\ -21983 & -127884 & 203906 & 178385 & 172193 & 311351 & -351663 & -329851 \end{matrix}$
146	$(5,992;0)$	$w_i = \dfrac{2003585969688162 73}{55307525215722359193600000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -125428 & 49546 & 264603 & -188721 & 264603 & -188721 & -403778 & 327896 \\ 125428 & -49546 & -264603 & 188721 & -264603 & 188721 & 403778 & -327896 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
147	$(5,992;1)$	$w_i = \dfrac{83453358234594783 77}{67208567603591162581920000000000000000}$
		$\begin{matrix} -90138 & 187114 & -220284 & 291484 & -220284 & 291484 & -26396 & -295050 \\ 91439 & -360183 & 41146 & 79400 & 41146 & 79400 & 160941 & -51221 \\ 91439 & -360183 & 41146 & 79400 & 41146 & 79400 & 160941 & -51221 \\ -116645 & 408110 & 1743 & -164990 & 1743 & -164990 & 914 & -47956 \end{matrix}$

i	$(m,b;\ell)$	w_i and f_i
148	$(5,992;1)$	$w_i = \frac{506456769176658779}{4498010717959953833120000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 55325 & 207438 & -54511 & -363169 & -54511 & -363169 & 185682 & 386916 \\ -55325 & -207438 & 54511 & 363169 & 54511 & 363169 & -185682 & -386916 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$
149	$(5,103;0)$	$w_i = \frac{8977039161301673}{43020226616174764800000000000000000}$
		$\begin{matrix} -20116 & -43919 & -43919 & 149754 & -125299 & 144600 & 120594 & -372072 \\ 168797 & -245624 & -245624 & 89335 & -135626 & 95882 & 90816 & 118272 \\ -125299 & 120594 & 144600 & -372072 & 52222 & -163150 & -163150 & 443421 \\ -135626 & 90816 & 95882 & 118272 & 160580 & -91105 & -91105 & -205858 \end{matrix}$
150	$(5,103;0)$	$w_i = \frac{828950085459354839}{31970605515319088049920000000000000000}$
		$\begin{matrix} 9489 & 11799 & 11799 & 97240 & 191567 & -410714 & -44468 & 63807 \\ 1704 & -98301 & -98301 & -5252 & -198499 & 434084 & 57229 & -115938 \\ 191567 & -44468 & -410714 & 63807 & -102654 & 94885 & 94885 & 89759 \\ -198499 & 57229 & 434084 & -115938 & 46797 & -78057 & -78057 & -88779 \end{matrix}$
151	$(5,628;0)$	$w_i = \frac{1782004166473368917}{24526725463748615520000000000000000000}$
		$\begin{matrix} -9988 & -56783 & 10458 & -37236 & -2453 & -114884 & -125757 & 183622 \\ -2453 & -114884 & -125757 & 183622 & -199555 & 347979 & 275795 & -451043 \\ -67848 & 260700 & -202107 & -76907 & 47712 & -237056 & 196963 & -8643 \\ 47712 & -237056 & 196963 & -8643 & 5383 & 92377 & -285274 & 159179 \end{matrix}$
152	$(5,628;1)$	$w_i = \frac{46769778056333567}{70169300037843706875000000000000000}$
		$\begin{matrix} 31522 & 143510 & -166495 & 111854 & 1766 & -71297 & 300591 & -188280 \\ 1766 & -71297 & 300591 & -188280 & -76659 & -25517 & -326825 & 223051 \\ 114540 & 187494 & -284195 & 204870 & -94584 & -166434 & 109695 & -234556 \\ -94584 & -166434 & 109695 & -234556 & 111095 & 175112 & -37918 & 300760 \end{matrix}$
153	$(5,433;0)$	$w_i = \frac{2755766554090241463}{27643309784734467430400000000000000000}$
		$\begin{matrix} 25911 & -82242 & -82242 & 146360 & 36794 & -108 & -108 & -216611 \\ 123834 & -105012 & -105012 & -67540 & 22046 & -166318 & -166318 & 408317 \\ 116477 & -143725 & -143725 & 17244 & -144662 & 18461 & 18461 & 203770 \\ -258725 & 151236 & 151236 & 26828 & -66769 & 243705 & 243705 & -544673 \end{matrix}$
154	$(5,613;0)$	$w_i = \frac{64450163453280917}{12563784648287565030400000000000000000}$
		$\begin{matrix} 5664 & -52839 & -52839 & 69922 & -52839 & 212699 & 69922 & -109522 \\ -52839 & 212699 & 212699 & -266068 & 69922 & -266068 & -109522 & 228938 \\ -52839 & 69922 & 212699 & -109522 & 212699 & -266068 & -266068 & 228938 \\ 69922 & -109522 & -266068 & 228938 & -109522 & 228938 & 228938 & -243640 \end{matrix}$
155	$(5,729;0)$	$w_i = \frac{4353630263669553}{63265668438845521600000000000000000}$
		$\begin{matrix} -84939 & -29650 & -29650 & 117316 & 17092 & 202526 & 202526 & -272757 \\ -29650 & 117316 & 117316 & -74296 & 202526 & -272757 & -272757 & 253325 \\ -29650 & 117316 & 117316 & -74296 & 202526 & -272757 & -272757 & 253325 \\ 117316 & -74296 & -74296 & -41495 & -272757 & 253325 & 253325 & -124322 \end{matrix}$
156	$(5,437;0)$	$w_i = \frac{2185743559064587}{35176179233161900000000000000000000}$
		$\begin{matrix} 0 & -61194 & 61194 & 0 & 0 & 172460 & -172460 & 0 \\ 0 & 30255 & -30255 & 0 & 0 & -141521 & 141521 & 0 \\ 0 & 288558 & -288558 & 0 & 0 & -399824 & 399824 & 0 \\ 0 & -257620 & 257620 & 0 & 0 & 368885 & -368885 & 0 \end{matrix}$
157	$(5,437;1)$	$w_i = \frac{17207575118691417}{21339370413697321025000000000000000000}$
		$\begin{matrix} 0 & -88947 & 88947 & 0 & 0 & 84338 & -84338 & 0 \\ 0 & 423517 & -423517 & 0 & 0 & -424400 & 424400 & 0 \\ 0 & -176671 & 176671 & 0 & 0 & 109860 & -109860 & 0 \\ 0 & -163346 & 163346 & 0 & 0 & 235649 & -235649 & 0 \end{matrix}$
158	$(5,437;1)$	$w_i = \frac{10185159341417010557}{65879231552397806527200000000000000000}$
		$\begin{matrix} -67920 & -71529 & -71529 & 330881 & -221282 & 149793 & 149793 & -129611 \\ 83933 & -716 & -716 & -145769 & 223608 & -225215 & -225215 & 221495 \\ -44272 & 26039 & 26039 & 47729 & -151004 & 26552 & 26552 & -26231 \\ -354393 & 123477 & 123477 & -4732 & 474104 & 28824 & 28824 & -350988 \end{matrix}$

i	$(m,b;\ell)$	w_i and f_i
159	$(5,945;0)$	$w_i = \dfrac{200607639057790431}{1834033434879661629440000000000000000}$
		$-102001\ \ 53022\ \ 53022\ -51522\ \ 57213\ \ 61967\ \ 61967\ -210729$
		$-123460\ \ 96065\ \ 96065\ -86483\ \ 161831\ -269930\ -269930\ \ 370788$
		$\ \ 292293\ -197472\ -197472\ \ 84837\ -227619\ \ -14844\ \ -14844\ \ 249307$
		$-127755\ \ 32246\ \ 32246\ \ 43873\ \ -7563\ \ 213132\ \ 213132\ -423227$
160	$(5,945;1)$	$w_i = \dfrac{2111976310256997093}{3214419668485395717580000000000000000}$
		$\ \ 38596\ \ -8311\ \ -8311\ -422884\ \ -69277\ -116414\ -116414\ \ 58650$
		$-231681\ \ 20131\ \ 20131\ \ 68569\ \ 15619\ \ 401321\ \ 401321\ \ -51045$
		$-146864\ 305857\ 305857\ \ 89406\ \ 132021\ -105372\ -105372\ \ 168833$
		$\ \ 106129\ -62501\ -62501\ \ -11623\ -117282\ -161972\ -161972\ -172646$
161	$(5,993;0)$	$w_i = \dfrac{2764895030026724011}{513615251231758033920000000000000000000}$
		$\ \ 45911\ \ 47720\ -132301\ \ -18897\ -132301\ \ -18897\ \ 232013\ \ \ 71419$
		$-132301\ -18897\ \ 173706\ \ 115666\ \ 173706\ \ 115666\ -132829\ -262849$
		$-132301\ -18897\ \ 173706\ \ 115666\ \ 173706\ \ 115666\ -132829\ -262849$
		$\ \ 232013\ \ 71419\ -132829\ -262849\ -132829\ -262849\ \ -16768\ \ 533750$
162	$(5,621;1)$	$w_i = \dfrac{2832231999020026609}{44794108197882542748400000000000000000}$
		$\ \ \ \ 0\ \ \ \ \ 0\ \ 188843\ -217853\ \ 157511\ -304729\ -276231\ -29171$
		$-157511\ 304729\ \ \ -3334\ \ -12126\ \ \ \ \ 0\ \ \ \ \ 0\ \ 120448\ 229425$
		$-188843\ 217853\ \ \ \ \ 0\ \ \ \ \ 0\ \ \ 3334\ \ \ 12126\ \ 390624\ \ 46536$
		$\ \ 276231\ \ 29171\ -390624\ \ -46536\ -120448\ -229425\ \ \ \ \ 0\ \ \ \ \ 0$
163	$(5,222;1)$	$w_i = \dfrac{16513606528458395 01}{41417733226532722220000000000000000000}$
		$0\ \ \ 66398\ -22739\ \ \ 14682\ -43659\ \ \ 7647\ \ -22329\ 0$
		$0\ -230639\ \ 78984\ -104605\ \ 151654\ -54480\ \ \ 159086\ 0$
		$0\ -230639\ \ 78984\ -104605\ \ 151654\ -54480\ \ \ 159086\ 0$
		$0\ \ \ -4532\ \ \ 1552\ \ \ 457158\ \ \ 2980\ \ 238097\ -695255\ 0$
164	$(5,1016;1)$	$w_i = \dfrac{339238114973793401}{30594658058681922000000000000000000000}$
		$\ \ \ \ 0\ \ \ \ \ 0\ \ \ \ \ 0\ \ \ \ \ 0\ -147301\ \ -26258\ \ -26258\ \ 137250$
		$-169467\ -30209\ -30209\ \ 157903\ \ 285513\ \ 202209\ \ 202209\ -555383$
		$\ \ 316769\ \ 56466\ \ 56466\ -295153\ -152746\ -108180\ -108180\ \ 297123$
		$-132767\ -94030\ -94030\ \ 258260\ \ \ \ \ 0\ \ \ \ \ 0\ \ \ \ \ 0\ \ \ \ \ 0$
165	$(5,757;0)$	$w_i = \dfrac{1497456138086954391}{23447468157191212825600000000000000000}$
		$-102286\ \ 193340\ \ 193340\ -252751\ \ 113756\ \ -37287\ \ -78441\ \ 125467$
		$\ \ -53433\ \ \ 44483\ \ \ 44483\ \ \ 63629\ \ 191996\ -219766\ -155037\ \ 133837$
		$\ \ 113756\ \ -78441\ \ -37287\ \ 125467\ \ \ 25287\ \ -72988\ \ -72988\ \ \ 48904$
		$\ \ 191996\ -155037\ -219766\ \ 133837\ -325934\ \ 376647\ \ 376647\ -331241$
166	$(5,757;1)$	$w_i = \dfrac{12778292139946942907}{29717211422188802648800000000000000000}$
		$\ \ \ \ 0\ \ 150522\ -150522\ \ \ \ \ 0\ 190124\ -168711\ -220907\ \ 15809$
		$\ \ \ \ 0\ -199609\ \ 199609\ \ \ \ \ 0\ \ 73444\ \ -11165\ -300224\ 421630$
		$-190124\ \ 220907\ \ 168711\ \ -15809\ \ \ \ \ 0\ \ -83906\ \ \ 83906\ \ \ \ \ 0$
		$\ \ -73444\ \ 300224\ \ \ 11165\ -421630\ \ \ \ \ 0\ -208262\ \ 208262\ \ \ \ \ 0$
167	$(5,757;1)$	$w_i = \dfrac{1821702848961542451}{77907580931444587346600000000000000000}$
		$\ \ 13689\ -240535\ -240535\ \ \ 47022\ \ \ 14064\ 364049\ -19802\ \ \ 59182$
		$\ \ 24393\ \ 191595\ \ 191595\ \ -17059\ -238548\ -27782\ \ 34466\ -155793$
		$\ \ 14064\ \ -19802\ \ 364049\ \ \ 59182\ -382421\ \ \ 2350\ \ \ 2350\ \ -36908$
		$-238548\ \ \ 34466\ \ -27782\ -155793\ \ 349408\ 139558\ 139558\ -243732$
168	$(5,1017;1)$	$w_i = \dfrac{3015401203760311}{15836595438435636000000000000000000}$
		$\ \ 59906\ \ 20837\ \ 20837\ \ \ 49577\ -178589\ -194589\ -194589\ \ 334640$
		$\ -13778\ \ 15824\ \ 15824\ -280444\ \ \ 38627\ \ 187994\ \ 187994\ \ -70070$
		$\ -13778\ \ 15824\ \ 15824\ -280444\ \ \ 38627\ \ 187994\ \ 187994\ \ -70070$
		$-251194\ 162318\ 162318\ \ 300548\ -159737\ \ \ 83714\ \ \ 83714\ -463653$
169	$(5,1005;1)$	$w_i = \dfrac{9678446144470417}{47417889785226202000000000000000000}$
		$-159012\ \ 120790\ \ 138044\ \ -76273\ \ 138044\ \ -76273\ \ 191233\ -179404$
		$\ \ 138044\ \ -76273\ \ -93359\ \ -14317\ \ 191233\ -179404\ -155451\ \ \ 92378$
		$\ \ 138044\ \ -76273\ \ 191233\ -179404\ \ -93359\ \ -14317\ -155451\ \ \ 92378$
		$\ \ 191233\ -179404\ -155451\ \ \ 92378\ -155451\ \ \ 92378\ -349576\ \ 561040$

i	$(m,b;\ell)$	w_i and f_i
170	$(5,1005;1)$	$w_i = \frac{9383199707071869}{592101906199158625000000000000000000}$
		$\begin{matrix} -115546 & -346863 & 174 & 247803 & 174 & 247803 & 43300 & 6275 \\ 174 & 247803 & -12047 & -5349 & 43300 & 6275 & -96333 & -266942 \\ 174 & 247803 & 43300 & 6275 & -12047 & -5349 & -96333 & -266942 \\ 43300 & 6275 & -96333 & -266942 & -96333 & -266942 & 351077 & 409019 \end{matrix}$
171	$(5,1021;0)$	$w_i = \frac{500668015173637}{201291205418100000000000000000000000}$
		$\begin{matrix} -15789 & 36646 & 36646 & -69653 & -53279 & 82915 & 82915 & -7871 \\ 36646 & -69653 & -69653 & 192826 & 82915 & -7871 & -7871 & -126949 \\ -53279 & 82915 & 82915 & -7871 & 393183 & -380782 & -380782 & 294544 \\ 82915 & -7871 & -7871 & -126949 & -380782 & 294544 & 294544 & -120862 \end{matrix}$
172	$(5,1021;1)$	$w_i = \frac{46617596546400257}{239048324379393750000000000000000000}$
		$\begin{matrix} 44964 & 67039 & 67039 & -104169 & -296156 & 23986 & 23986 & -41206 \\ 67039 & -104169 & -104169 & 94276 & 23986 & -41206 & -41206 & 319968 \\ -296156 & 23986 & 23986 & -41206 & 171838 & 293153 & 293153 & -254235 \\ 23986 & -41206 & -41206 & 319968 & 293153 & -254235 & -254235 & -260743 \end{matrix}$
173	$(5,1021;1)$	$w_i = \frac{5156467371309277}{399189544161133359375000000000000000}$
		$\begin{matrix} 0 & 0 & 0 & 0 & -98594 & -112503 & -112503 & 292684 \\ 0 & 0 & 0 & 0 & -112503 & 292684 & 292684 & -441947 \\ 98594 & 112503 & 112503 & -292684 & 0 & 0 & 0 & 0 \\ 112503 & -292684 & -292684 & 441947 & 0 & 0 & 0 & 0 \end{matrix}$
174	$(5,1023;1)$	$w_i = \frac{32537947759658797}{169761371750002687500000000000000000}$
		$\begin{matrix} -79328 & -281850 & -281850 & 195844 & -281850 & 195844 & 195844 & 54893 \\ -281850 & 195844 & 195844 & 54893 & 195844 & 54893 & 54893 & -187908 \\ -281850 & 195844 & 195844 & 54893 & 195844 & 54893 & 54893 & -187908 \\ 195844 & 54893 & 54893 & -187908 & 54893 & -187908 & -187908 & -79251 \end{matrix}$

Certificate package

The accompanying `2026-10-14` local certificate snapshot is `srg-gegenbauer-counting.zip`. Within `results/certificates/405_132_63_33/`, its exact input is `ordinary_counting_certificate/exact_certificate.json`. The common command `python3 -S code/verify.py` checks this file against the printed tables using Python's standard library. The input manifest, release hashes, and full verification inventory are specified in Appendix A of the main article and in `docs/reproducibility_snapshot.md`.

S2 The all-degree five-point relaxation for 405

The same parameters admit a rational feasible point for the five-point real-count relaxation at *every* Gegenbauer degree. Temporarily put $q_F = N_F/(405|\operatorname{Aut} F|)$ and choose

F	$K_5 \setminus P_3$	$K_5 \setminus 2K_2$	$K_5 \setminus K_2$	K_5
q_F	988060	134777	287429	68038

The first graph has two adjacent edges deleted. Exact elimination reconstructs all 52 nonnegative counts and verifies the ordinary extension equations. All profile matrices with at most three reference vertices are PSD at every degree. The accompanying all-degree verifier checks the 66 necessary low-degree matrices by exact Schur elimination; the remaining degrees follow from the rational tail estimate described below. This is a feasible point of a real relaxation, not a graph construction.

Here $P_\ell^{(D)}$ is the Gegenbauer polynomial for S^{D-1}, normalized by $P_\ell^{(D)}(1) = 1$. At every degree the profile matrices use the extension coefficients of Section S1 and the homogeneous kernel $\kappa_\ell(z, h, h') = (hh')^{\ell/2} P_\ell^{(D)}(z/\sqrt{hh'})$. For $q \in \mathbb{Z}_{\geq 0}$ and $D > 2$, put

$$B_{q,D}(s) = \sum_{j=0}^{q} \binom{q}{j} s^{q-j}(1-s)^j \prod_{a=1}^{j} \frac{2a-1}{D+2a-3},$$

with the empty product equal to one. The normalized Gegenbauer integral representation [DLMF, 18.10.4] gives $|P_\ell^{(D)}(t)| \leq B_{q,D}(s)$ whenever $t^2 \leq s < 1$ and $\ell \geq 2q$. Indeed, $P_\ell^{(D)}(t) = \mathbb{E}(t + i\sqrt{1-t^2}Z)^\ell$, where Z is the first coordinate of a uniform point on S^{D-2}; take absolute values and use $\mathbb{E}Z^{2j} = \prod_{a=1}^{j}(2a-1)/(D+2a-3)$. The expression $t^2 + (1-t^2)Z^2$ is at most one and increases with t^2, which proves the bound. For an anchor with $1 \leq m \leq 3$, undo the common positive normalization of Section S1, which preserves positive semidefiniteness. Fix its reference graph H, put $D = 30 - m$, and enumerate its profiles as $\sigma_1, \ldots, \sigma_{2^m}$ in binary lexicographic order. Put $n_i = N_{H_{\sigma_i}}$ and $c_{ij}(e) = N_{H_{\sigma_i, \sigma_j, e}}$. Swapping the two new labels gives $c_{ij}(e) = c_{ji}(e)$. These nonnegative single-profile and distinct-pair masses obey $\sum_{j,e} c_{ij}(e) = (404 - m)n_i$. After dividing coordinate i by $h_{\sigma_i}^{\ell/2}$, the diagonal contribution is $\operatorname{diag}(n_i)$. For any real column $u = (u_i)_{i=1}^{2^m}$, the absolute quadratic contribution of all distinct pairs is at most $(404 - m)B_{q,D}(s)\sum_i n_i u_i^2$, by $2|u_i u_j| \leq u_i^2 + u_j^2$. Thus every tail matrix is PSD if $(404 - m)B_{q,D}(s) \leq 1$.

The exact maximum squared residual angles and sufficient tail cutoffs for the seven nonempty anchor types are

H	D	s	ℓ in the tail
K_1	29	3/35	$\ell \geq 8$
$\overline{K_2}$	28	1/8	$\ell \geq 8$
K_2	28	9/64	$\ell \geq 8$
$\overline{K_3}$	27	1/4	$\ell \geq 10$
$K_2 \dot\cup K_1$	27	867/4715	$\ell \geq 10$
P_3	27	3/11	$\ell \geq 12$
K_3	27	4/21	$\ell \geq 10$

All heights are positive, and the displayed tail inequalities are checked over $\mathbb{Q}$. Hence increasing the degree alone cannot exclude 405 within this five-point real-count relaxation, whereas seven points and degrees at most two suffice. The sufficiency of the intermediate six-point level is not determined by these results.

References

[DLMF] NIST Digital Library of Mathematical Functions, Section 18.10(ii), equation 18.10.4, Laplace-type integral representation for ultraspherical polynomials.